

How Much Can We Compress Neural Training Dynamics?

Response-memory closures for deep nonlinear feature learning

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Abstract

Compressing a trained function is easier than compressing its training dynamics: two parameter states can define the same input–output map yet have different tangent features and separate as soon as training resumes. We study a more demanding target: an autonomous state whose size does not grow with elapsed training time and whose trajectory approximates gradient flow on every prescribed finite time interval. The setting keeps fixed depth, nonlinear activations and hidden-feature learning under the canonical width-dependent mobilities.

The exact learned part of every hidden matrix is an integral of a backward response against a forward response. Exact history representations retain all past responses and therefore do not compress this integral in time. Our closure stores only the first P shifted-Legendre moments of each response history, while retaining the initialized matrices exactly. Moment differentiation closes because the derivative of a Legendre mode is triangular in lower modes. An ordinary residual-activity clock gives a fully autonomous closure with compact-horizon trajectory error $O_T(P^{-1})$. A response-speed clock, which parametrizes the joint forward/backward response path by residual mass plus path length, gives $O_T(P^{-2})$ under one additional activation derivative. Both statements hold at fixed finite width and data for arbitrary fixed depth; they require neither successful fitting nor special input geometry. The second rate is an asymptotic guarantee, not a claim of lower practical cost at small order.

The moving learned state falls from $(H - 1)n^2$ dense entries to $2(H - 1)mnP$ history entries, but every initialized $n \times n$ matrix and its forward/transpose actions remain. Existing circle and MNIST experiments show that very small orders can reproduce the dense fitted predictor, including on passive inputs. The report also separates this result from a width-uniform population theorem, unrestricted scalar nonclosure, all-time uniform tracking, and full compression of the initialized random operator, none of which is proved here.

1 The question is dynamical, not only functional

The primary question is simple to state: *how much can the training dynamics of a neural network be compressed?* A snapshot compressor need only reproduce the current function. A training compressor must also reproduce how the function will move. Equality of predictions does not identify the parameter-space Jacobian, the backward responses, or the optimizer metric. Consequently, two exact functional copies can acquire different velocities and their errors can accumulate after training continues.

This stricter question can be scientifically useful. A successful compression identifies a smaller state that retains the mechanism responsible for learning. It may therefore offer a better set of coordinates for studying deep learning, rather than only a smaller file format for a trained predictor.

We impose three non-negotiable features: more than one hidden layer, a nonlinearity that remains active during the limiting description, and motion of hidden representations at leading order. The canonical mobilities used below are of the same feature-learning type emphasized in

tensor-program and mean-field work [Yang and Hu, 2021, Nguyen and Pham, 2023]. Population descriptions are attractive because width becomes a particle approximation and many macroscopic observables can become deterministic. They do not automatically become finite-dimensional: reused random matrices can survive as operators, and exact descriptions may retain entire one- or two-time response histories. Such a description organizes the limit, but it does not by itself provide state whose size is constant in training time. Particle and measure gradient flows provide an important shallow precedent [Chizat and Bach, 2018].

The motivating analogy is with statistical physics: more particles can permit a smoother field description even while the microscopic system grows. A more speculative analogy is with models of computation. A well-posed population flow might play the role of an idealized model of learning, while finite networks approximate it with finite width, finite precision and finite time steps. These analogies motivate the problem; they are not mathematical claims of this report.

1.1 Claim levels and state classes

We use the following distinctions throughout.

Definition 1.1 (Compact-horizon global approximation). Let $z(t)$ be a reference flow and $z_P(t)$ an autonomous order- P closure. The closure is *globally accurate on compact horizons* if, for every fixed $T < \infty$,

$$\sup_{0 \leq t \leq T} d(z_P(t), z(t)) \longrightarrow 0 \quad (P \rightarrow \infty).$$

The order may depend on T . This does not mean that one fixed order works for all $t \geq 0$, nor that the constants remain bounded as $T \rightarrow \infty$.

Three kinds of autonomous state are relevant.

1. A *scalar ODE* evolves finitely many real numbers. Its dimension is independent of width, training time and any discretization of a population.
2. A *population or field ODE* evolves finitely many functions, random variables or densities on fixed source spaces. It has finitely many fields but infinitely many scalar degrees of freedom.
3. An *operator ODE* evolves maps between population spaces. A single operator retains the row-column incidence that a finite collection of marginal populations may lose. “One operator” therefore need not mean low information.

Exact and approximate closure are separate questions in each class. So are a local velocity match and uniform trajectory control on $[0, T]$.

Two established comparison results help locate the present contribution. First, for a one-input, three-hidden-layer linear feature-learning system, companion theory constructs a fixed Hilbert space, a bounded source $\mathcal{C}_0$ and a unique global trace-class increment Q satisfying

$$\mathcal{C} = \mathcal{C}_0 + Q, \quad f = \frac{1}{4} \text{Tr } \mathcal{C}^4, \quad \dot{Q} = -2\kappa(f - y)(\mathcal{C}_0^* + Q^*)^3.$$

Finite gradient flow, and raw gradient descent for every step sequence tending to zero, converge to its observables on compact horizons. This is a scoped example of exact operator closure, not a general deep nonlinear theorem.

Second, in the same benchmark there is no state-universal, width-uniform encoder built from a fixed finite alphabet of bounded-degree contraction graphs that has a polynomial autonomous scalar

ODE (or a finite-order local polynomial PDE) and a polynomial readout. The proof differentiates the output repeatedly: the $(2j - 1)$ st derivative contains a connected contraction

$$u^\top (B^\top R^\top RB)^j u$$

of degree $4j + 2$. A fixed bounded-degree alphabet cannot generate these pairwise distinct connected graphs. This is evidence against simple exact scalar closures. It is not an impossibility theorem for arbitrary encoders; indeed an unrestricted field can encode a whole future output profile. We therefore treat broad scalar impossibility and exact operator-to-population compression as open questions rather than conclusions.

2 Dense deep gradient flow

Fix a dataset $(x_a, y_a)_{a=1}^m$, $x_a \in \mathbb{R}^d$, a width n , and $H \geq 2$ hidden layers. Set $v_a = x_a/\sqrt{d}$. For simplicity every hidden layer has width n ; activations may differ by layer:

$$\begin{aligned} z_a^{(1)} &= W^{(1)} v_a, & h_a^{(1)} &= \phi_1(z_a^{(1)}), \\ z_a^{(\ell)} &= W^{(\ell)} h_a^{(\ell-1)}, & h_a^{(\ell)} &= \phi_\ell(z_a^{(\ell)}), \quad 2 \leq \ell \leq H, \\ f_a &= \frac{w^\top h_a^{(H)}}{n}, & r_a &= f_a - y_a, \quad \rho = \left(\frac{1}{m} \sum_a r_a^2 \right)^{1/2}. \end{aligned} \tag{1}$$

The loss is $\mathcal{L} = \rho^2$. Backward responses exclude the residual:

$$\begin{aligned} \delta_a^{(H)} &= w \odot \phi'_H(z_a^{(H)}), \\ \delta_a^{(\ell)} &= \phi'_\ell(z_a^{(\ell)}) \odot (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}. \end{aligned} \tag{2}$$

With mobilities $(n, 1, \dots, 1, n)$, the physical gradient field F is

$$\begin{aligned} F_1 &= -\frac{2}{m} \sum_a r_a \delta_a^{(1)} v_a^\top, \\ F_\ell &= -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top, \quad 2 \leq \ell \leq H, \\ F_w &= -\frac{2}{m} \sum_a r_a h_a^{(H)}. \end{aligned} \tag{3}$$

Thus every internal matrix has the exact representation

$$W^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t r_a(s) \delta_a^{(\ell)}(s) (h_a^{(\ell-1)}(s))^\top ds. \tag{4}$$

The growing history in (4) is the object we compress. The initialized matrices $W_0^{(\ell)}$ are retained exactly.

3 Variant A: the residual-activity clock

Let p_k be shifted Legendre polynomials on $[0, 1]$, normalized by $p_k(1) = 1$ and

$$\int_0^1 p_j(x) p_k(x) dx = \frac{\mathbf{1}_{j=k}}{2k+1}.$$

They satisfy the triangular identity

$$xp'_k(x) = kp_k(x) + \sum_{j < k} (2j + 1)p_j(x). \quad (5)$$

Define the activity coordinate

$$\tau(t) = 1 + \int_0^t \rho(s) ds. \quad (6)$$

At internal link ℓ and sample a , define

$$a_{\ell a} = h_a^{(\ell-1)}, \quad b_{\ell a} = \frac{r_a}{\rho} \delta_a^{(\ell)} \quad (\rho > 0).$$

In the coordinate $\xi = \tau(s)$, extend $a_{\ell a}$ constantly over the artificial prefix $[0, 1]$ and set $b_{\ell a} = 0$ there. Store only

$$H_{\ell ak} = \int_0^\tau \bar{a}_{\ell a}(\xi) p_k(\xi/\tau) d\xi, \quad U_{\ell ak} = \int_0^\tau \bar{b}_{\ell a}(\xi) p_k(\xi/\tau) d\xi, \quad 0 \leq k < P. \quad (7)$$

Differentiating and using (5) gives the closed ODE

$$\begin{aligned} \dot{H}_{\ell ak} &= \rho a_{\ell a} - \frac{\rho}{\tau} \left[k H_{\ell ak} + \sum_{j < k} (2j + 1) H_{\ell aj} \right], \\ \dot{U}_{\ell ak} &= r_a \delta_a^{(\ell)} - \frac{\rho}{\tau} \left[k U_{\ell ak} + \sum_{j < k} (2j + 1) U_{\ell aj} \right]. \end{aligned} \quad (8)$$

Initially $H_{\ell a0} = a_{\ell a}(0)$ and all other moments vanish. The reconstructed matrices are

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm\tau} \sum_{a=1}^m \sum_{k=0}^{P-1} (2k + 1) U_{\ell ak} H_{\ell ak}^\top. \quad (9)$$

All responses in (8) are evaluated in the current reconstructed network. The first matrix and readout evolve by their exact blocks in (3). Equations (6)–(9) are therefore an autonomous finite ODE. They never divide by ρ ; the normalized b is only a proof coordinate.

If Π_P denotes componentwise $L^2(0, \tau)$ projection onto polynomials of degree less than P , then (9) is exactly

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^\tau (\Pi_P \bar{b}_{\ell a})(\xi) (\Pi_P \bar{a}_{\ell a})(\xi)^\top d\xi. \quad (10)$$

Thus closure of the moment ODE is exact; truncating the two history projections is the modeling approximation.

3.1 Compact-horizon convergence

Let $\theta = (W^{(1)}, \dots, W^{(H)}, w)$ and use the Euclidean block norm, with Frobenius norms on matrices.

Theorem 3.1 (Ordinary-clock response-memory convergence). *Fix finite n, d, m, H , finite data and finite initialization. Suppose $\phi_\ell \in C_{\text{loc}}^{1,1}(\mathbb{R})$ for every layer. For every $T < \infty$ there are finite*

constants $C_{\text{old}}(T)$ and $P_0(T)$, independent of P , such that the order- P closure exists uniquely through T for every $P \geq P_0(T)$ and

$$\sup_{0 \leq t \leq T} \left\| \widehat{\theta}_P(t) - \theta(t) \right\| \leq \frac{C_{\text{old}}(T)}{\sqrt{P(P+1)}}. \quad (11)$$

The same rate holds for all hidden responses and predictions on the training set and on any fixed bounded collection of passive inputs.

If, in addition, every ϕ_ℓ and ϕ'_ℓ is globally bounded, then the closure exists uniquely for all finite physical times for every $P \geq 1$, and (11) holds for every $P \geq 1$ after enlarging the constant. This includes $\tanh$, logistic sigmoid, arctangent and smooth bounded periodic activations. No fitting, Gaussian initialization, input separation or label compatibility assumption is required.

The word “global” in this theorem means the whole interval $[0, T]$ for each arbitrary finite T . It does not give a T -independent constant.

3.2 Proof of the ordinary-clock theorem

We give the proof because it shows exactly where history regularity enters and why feedback does not invalidate the projection estimate.

Lemma 3.2 (Legendre tail). *For $q \in H^1(0, \tau; \mathbb{R}^N)$,*

$$\|q - \Pi_P q\|_{L^2(0, \tau)}^2 \leq \frac{1}{P(P+1)} \int_0^\tau \xi(\tau - \xi) \|q'(\xi)\|^2 d\xi \leq \frac{\tau^2}{4P(P+1)} \|q'\|_{L^2(0, \tau)}^2. \quad (12)$$

Proof. On $[0, 1]$, the shifted Legendre equation is

$$-\frac{d}{dx} \{x(1-x)p'_k(x)\} = k(k+1)p_k(x).$$

Integration by parts shows that the derivatives are orthogonal in the weight $x(1-x)$ and have squared norm $k(k+1)/(2k+1)$. Bessel's inequality applied to q' in this weighted space gives

$$\sum_{k \geq 1} k(k+1) \frac{\|c_k\|^2}{2k+1} \leq \int_0^1 x(1-x) \|q'(x)\|^2 dx,$$

where $c_k = (2k+1) \int_0^1 q(x)p_k(x) dx$. Parseval, polynomial density and $k(k+1) \geq P(P+1)$ for $k \geq P$ give the first bound on $[0, 1]$. Rescaling and $\xi(\tau - \xi) \leq \tau^2/4$ give (12); summing componentwise proves the vector statement. $\square$

Step 1: dense existence and a common compact region. The mobility matrix $\mathcal{D}$ has largest eigenvalue n , and $F = -\mathcal{D}\nabla\mathcal{L}$. Hence

$$\dot{\mathcal{L}} = -\left\| \mathcal{D}^{-1/2} \dot{\theta} \right\|^2, \quad \int_0^t \left\| \dot{\theta}(s) \right\|^2 ds \leq n\rho(0)^2. \quad (13)$$

Thus $\|\theta(t) - \theta(0)\| \leq \sqrt{nt}\rho(0)$. A finite maximal endpoint would have a finite limiting state, from which local existence extends the solution. Dense flow is therefore global.

Fix T and let

$$R = \|\theta(0)\| + \sqrt{nT}\rho(0) + 1, \quad \mathcal{B} = \{\vartheta : \|\vartheta\| \leq R\}.$$

The dense path remains distance at least one from $\partial\mathcal{B}$. Stop the closure at its first exit from $\mathcal{B}$. The rest of the proof uses constants on this explicit ball and will rule out that exit.

For a concrete choice of constants set $X = \max_a \|v_a\|$ and $\alpha_0 = X$. Recursively choose

$$\begin{aligned} s_\ell &= \sup_{|z| \leq R\alpha_{\ell-1}} |\phi'_\ell(z)|, & \alpha_\ell &= \sqrt{n} \sup_{|z| \leq R\alpha_{\ell-1}} |\phi_\ell(z)|, \\ \beta_H &= Rs_H, & \beta_\ell &= s_\ell R\beta_{\ell+1} \quad (\ell < H). \end{aligned} \quad (14)$$

Then $\|h_a^{(\ell)}\| \leq \alpha_\ell$ and $\|\delta_a^{(\ell)}\| \leq \beta_\ell$ on $\mathcal{B}$. Put

$$\begin{aligned} Y &= \left(m^{-1} \sum_a y_a^2 \right)^{1/2}, & q &= R\alpha_H/n + Y, & S &= Tq, & A &= 1 + S, \\ v_1 &= 2\beta_1 X, & v_\ell &= 2\beta_\ell \alpha_{\ell-1}/n, & v_w &= 2\alpha_H, \\ V &= \left(v_1^2 + \sum_{\ell=2}^H v_\ell^2 + v_w^2 \right)^{1/2}. \end{aligned} \quad (15)$$

On the stopped interval, $\rho \leq q$, $\tau \leq A$ and $\|F\| \leq V\rho$. Since F is locally Lipschitz, it has some finite Lipschitz constant K on $\mathcal{B}$.

Step 2: an exact projection-error energy. For any one forward or backward history q let $D_q(t) = \|q - \Pi_P q\|_{L^2(0, \tau(t))}^2$ and let q^* be the polynomial projection evaluated at the current endpoint. Differentiating the raw energy and the projected energy, using (8), cancels every basis-dilation term and gives

$$\dot{D}_q = \rho \|q - q^*\|^2, \quad D_q(t) = \int_0^t \rho(s) \|q(s) - q^*(s)\|^2 ds. \quad (16)$$

The backward-prefix jump is harmless: only square integrability and growing integrals are used.

Differentiate (9). The physical closure equation is exactly

$$\hat{\theta}_P = F(\hat{\theta}_P) + E, \quad E_1 = E_w = 0, \quad (17)$$

with internal defects

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_{\ell a} - b_{\ell a}^*)(a_{\ell a} - a_{\ell a}^*)^\top. \quad (18)$$

The two mixed projection terms vanish by orthogonality. Consequently

$$\int_0^t \|E_\ell(s)\|_F ds \leq \frac{2}{nm} \sum_a \sqrt{D_{b, \ell a}(t) D_{a, \ell a}(t)}. \quad (19)$$

This is the bridge from a same-history approximation to a trajectory theorem.

Step 3: forward histories have a P -independent derivative bound. Use primes for derivatives in $\xi = \tau(t)$ and define

$$Z_\ell(t) = \frac{1}{m} \sum_a \int_0^{\tau(t)} \left\| (h_a^{(\ell)})'(\xi) \right\|^2 d\xi.$$

The first layer satisfies $\|(h_a^{(1)})'\| \leq s_1 v_1 X$. For later layers, the chain rule gives

$$(h_a^{(\ell)})' = \phi_\ell'(z_a^{(\ell)}) \odot \left[(F_\ell/\rho + E_\ell/\rho) h_a^{(\ell-1)} + \widehat{W}^{(\ell)}(h_a^{(\ell-1)})' \right].$$

Endpoint evaluation of a degree- $(P-1)$ polynomial has operator norm $P/\sqrt{\tau}$ from $L^2(0, \tau)$, because $\sum_{k < P} (2k+1) = P^2$. Projection contraction and $m^{-1} \sum_a \|b_{\ell a}\|^2 \leq \beta_\ell^2$ imply

$$\left(m^{-1} \sum_a \|b_{\ell a}^*\|^2 \right)^{1/2} \leq P\beta_\ell, \quad \left(m^{-1} \sum_a \|b_{\ell a} - b_{\ell a}^*\|^2 \right)^{1/2} \leq (P+1)\beta_\ell.$$

Combining this with Lemma 3.2, (16) and (18) yields

$$\int_0^t \rho \|E_\ell/\rho\|_F^2 ds \leq \frac{2\beta_\ell^2 A^2}{n^2} Z_{\ell-1}(t). \quad (20)$$

The endpoint factor $P+1$ cancels the P^{-1} projection factor. This is the step that prevents a loss of convergence order with depth.

Define finite, P -independent bounds recursively by

$$\begin{aligned} \bar{Z}_1 &= S(s_1 v_1 X)^2, \\ \bar{Z}_\ell &= 3s_\ell^2 \left[S v_\ell^2 \alpha_{\ell-1}^2 + \left(R^2 + \frac{2\beta_\ell^2 A^2 \alpha_{\ell-1}^2}{n^2} \right) \bar{Z}_{\ell-1} \right]. \end{aligned} \quad (21)$$

Integrating (3.2), using $\|x+y+z\|^2 \leq 3(\|x\|^2 + \|y\|^2 + \|z\|^2)$ and (20), proves inductively that $Z_\ell \leq \bar{Z}_\ell$.

Step 4: accumulated defect and feedback. The zero backward prefix gives $m^{-1} \sum_a D_{b, \ell a} \leq S\beta_\ell^2$. By Lemma 3.2,

$$\frac{1}{m} \sum_a D_{a, \ell a} \leq \frac{A^2 \bar{Z}_{\ell-1}}{4P(P+1)}.$$

Summing (19) over links gives

$$\int_0^t \|E(s)\| ds \leq \frac{B_{\text{old}}}{\sqrt{P(P+1)}}, \quad B_{\text{old}} = \frac{A}{n} \sum_{\ell=2}^H \beta_\ell \sqrt{S \bar{Z}_{\ell-1}}. \quad (22)$$

Subtracting dense flow from the perturbed flow and applying the Lipschitz bound on F gives

$$e(t) \leq \frac{B_{\text{old}}}{\sqrt{P(P+1)}} + K \int_0^t e(s) ds.$$

Gronwall's inequality therefore yields

$$\sup_{t \leq T} e(t) \leq \frac{B_{\text{old}} e^{KT}}{\sqrt{P(P+1)}}. \quad (23)$$

Choose P_0 so the right side is less than $1/2$. The closure then cannot reach $\partial \mathcal{B}$. Its moment representations, $1 \leq \tau \leq A$ and the response bounds place its whole finite state in a compact subset of the locally Lipschitz domain. A finite maximal endpoint would therefore extend, so the

closure exists through T . Lipschitz dependence of network responses on θ gives the final prediction statement.

For completeness, suppose every activation and first derivative is globally bounded. Set

$$\alpha_\ell = \sqrt{n} \|\phi_\ell\|_\infty, \text{quad}s_\ell = \|\phi'_\ell\|_\infty, \text{quad}B_w = \sqrt{\|w(0)\|^2 + nY^2T}.$$

The unchanged readout equation obeys

$$\frac{d}{dt} \|w\|^2 = nY^2 - 4n \frac{1}{m} \sum_a (f_a - y_a/2)^2 \leq nY^2,$$

for both dense and closure flows. Therefore $\|w\| \leq B_w$, $\rho \leq q := B_w \alpha_H / n + Y$, $S = Tq$ and $\tau \leq A := 1 + S$. Start with $\beta_H = B_w s_H$ and proceed downward. For $\ell = H, \dots, 2$, define

$$D_\ell = \left\| W_0^{(\ell)} \right\|_F + \frac{2A\beta_\ell \alpha_{\ell-1}}{n}, \quad \beta_{\ell-1} = s_{\ell-1} D_\ell \beta_\ell.$$

Projection contraction in (10) gives $\left\| \widehat{W}^{(\ell)} \right\|_F \leq D_\ell$; the backward recurrence then gives $\|\delta^{(\ell-1)}\| \leq \beta_{\ell-1}$. There is no circularity because the descent starts from the bounded readout. Finally

$$\left\| W^{(1)}(t) \right\|_F \leq \left\| W^{(1)}(0) \right\|_F + 2SX\beta_1.$$

These P -independent physical bounds, the moment integral representations and $1 \leq \tau \leq A$ put the order- P state in a compact set for every fixed P . Hence the closure continues through T for every $P \geq 1$. Repeating Steps 2–4 with these global constants gives (23) for every P . This completes the proof.

4 Variant B: a joint response-speed clock

The activity clock controls the first-layer forward history particularly cleanly, but it does not make every backward history smooth. A more symmetric coordinate parametrizes the entire encoded response path. Concatenate

$$\Psi(\theta) = \text{stack}_{\ell=2, \dots, H; a=1, \dots, m}(a_{\ell a}, b_{\ell a}), \quad \dot{L} = g = \rho + \left\| \dot{\Psi} \right\|_2, \quad L(0) = 1. \quad (24)$$

Then $\|d\Psi/dL\| \leq 1$. The clock changes where history is placed, but the learning integral still carries residual mass. At time s , place a point at $\xi = L(s)$ and assign mass $\rho(s) ds$. Give $[0, 1]$ Lebesgue mass and matching constant prefixes. Thus

$$\int q d\mu_t = \int_0^1 q(\xi) d\xi + \int_0^t q(L(s))\rho(s) ds, \quad d\mu_t \leq d\xi.$$

The mass is $A(t) = 1 + \int_0^t \rho ds$, generally different from $L(t)$.

Let $p = (p_0, \dots, p_{P-1})^\top$, $e = (1, \dots, 1)^\top$, and let $\mathsf{T}_{kk} = k$, $\mathsf{T}_{kj} = 2j + 1$ for $j < k$. Store one shared Gram matrix and link/sample moments

$$H_{\ell a} = \int \bar{a}_{\ell a} p^\top d\mu, \quad U_{\ell a} = \int \bar{b}_{\ell a} p^\top d\mu, \quad G = \int p p^\top d\mu,$$

where p is evaluated at $\xi/L(t)$. Put $C_{\ell a} = b_{\ell a}(0)a_{\ell a}(0)^\top$. The reconstruction and exact moment equations are

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a (U_{\ell a} G^{-1} H_{\ell a}^\top - C_{\ell a}), \quad (25)$$

$$\begin{aligned} \dot{H}_{\ell a} &= \rho a_{\ell a} e^\top - (g/L) H_{\ell a} \mathsf{T}^\top, & \dot{U}_{\ell a} &= r_a \delta_a^{(\ell)} e^\top - (g/L) U_{\ell a} \mathsf{T}^\top, \\ \dot{G} &= \rho e e^\top - (g/L) (\mathsf{T} G + G \mathsf{T}^\top). \end{aligned} \quad (26)$$

Initially $H_{\ell a} = [a_{\ell a}(0), 0, \dots]$, $U_{\ell a} = [b_{\ell a}(0), 0, \dots]$ and $G = \text{diag}(1, (1/3), \dots, (1/(2P-1)))$. The prefix correction makes $\widehat{W}^{(\ell)}(0) = W_0^{(\ell)}$.

Solve $Gq = e$ and set $a_{\ell a}^* = H_{\ell a} q$ and $b_{\ell a}^* = U_{\ell a} q$. Product differentiation cancels every g/L term and gives the physical defect

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_{\ell a} - b_{\ell a}^*)(a_{\ell a} - a_{\ell a}^*)^\top. \quad (27)$$

This has the same product-of-errors form as Variant A, but now both factors belong to one path with unit speed in L .

Theorem 4.1 (Response-clock convergence). *Under the finite setup of Theorem 3.1, assume $\phi_1 \in C_{\text{loc}}^{1,1}$ and $\phi_\ell \in C_{\text{loc}}^{2,1}$ for $\ell \geq 2$. If $\rho(0) > 0$, then for every finite T there are $P_0(T)$, Λ_T and C_T , independent of P , such that for $P \geq P_0(T)$ the ODE (24)–(26) has a unique regular solution on $[0, T]$, $L_P(t) \leq \Lambda_T$, and*

$$\int_0^T \|E(t)\| \, dt \leq \frac{C_T}{P(P+1)}, \quad \sup_{t \leq T} \|\widehat{\theta}_P(t) - \theta(t)\| \leq \frac{C_T e^{K_T T}}{P(P+1)}. \quad (28)$$

If $\rho(0) = 0$, both flows are stationary. The conclusion is $O_T(P^{-2})$ at fixed width, depth and data. It is not an all-small-order or uniform-all-time statement.

Proof architecture. For a history q , let D_q be its weighted least-squares error. Differentiating $\text{tr}(HG^{-1}H^\top)$ with (26) proves the exact identity

$$\dot{D}_q = \rho \|q - q^*\|^2.$$

Consequently, Cauchy–Schwarz in physical time gives the analogue of (19). On a compact physical ball, raw history energies give a coarse, order-independent bound on $\int \|E\|$ and hence on total physical path variation. Dense residual satisfies a finite-horizon lower bound $\rho_{\text{dense}} \geq \mu_T > 0$. On the region $\rho \geq \mu_T/2$, the response map has a finite derivative bound $\|D\Psi\| \leq J_T$. Therefore

$$L_P(T) \leq A_T + J_T \left(\int_0^T \|F\| \, dt + \int_0^T \|E\| \, dt \right) =: \Lambda_T,$$

without assuming a clock bound.

The joint extended history is 1-Lipschitz in ξ and constant on the prefix. Weighted best approximation is no worse than ordinary Legendre projection because $d\mu \leq d\xi$. Applying Lemma 3.2 to the whole concatenated path yields

$$\sum_{\ell, a} (D_{a, \ell a} + D_{b, \ell a}) \leq \frac{L_P^2 (L_P - 1)}{4P(P+1)}.$$

The arithmetic–geometric mean inequality in the defect estimate gives

$$\int_0^T \|E\| \, dt \leq \frac{\Lambda_T^2(\Lambda_T - 1)}{4nmP(P + 1)}.$$

Gronwall gives (28). For sufficiently large P , this keeps the closure inside the physical ball and keeps its residual above $\mu_T/2$. The prefix makes G positive definite; bounded L gives a positive minimum eigenvalue for each fixed P . All moment variables then stay in a compact subset of the locally Lipschitz domain, which proves continuation through T and closes the first-exit argument. $\square$

4.1 The new clock is fully implementable

Equation (24) appears implicit only because $\dot{\Psi}$ is written compactly. No nonlinear solve for g is needed. First compute $q = G^{-1}e$, endpoint fits a^*, b^* , and internal velocities

$$V_\ell = \dot{\widehat{W}}^{(\ell)} = -\frac{2\rho}{nm} \sum_a \left[b_{\ell a} (a_{\ell a}^*)^\top + b_{\ell a}^* (a_{\ell a} - a_{\ell a}^*)^\top \right]. \quad (29)$$

These expressions contain no g . Compute $V_1 = F_1$ and $V_w = F_w$, then a directional forward pass

$$\dot{z}_a^{(1)} = V_1 v_a, \quad \dot{z}_a^{(\ell)} = V_\ell h_a^{(\ell-1)} + \widehat{W}^{(\ell)} \dot{h}_a^{(\ell-1)}, \quad \dot{h}_a^{(\ell)} = \phi'_\ell(z_a^{(\ell)}) \odot \dot{z}_a^{(\ell)}.$$

This gives $\dot{r}$ and $\dot{\rho}$. A reverse directional pass differentiates the backward recurrence, using ϕ''_ℓ for $\ell \geq 2$, and then

$$\dot{b}_{\ell a} = \frac{\dot{r}_a \delta_a^{(\ell)} + r_a \dot{\delta}_a^{(\ell)}}{\rho} - b_{\ell a} \frac{\dot{\rho}}{\rho}.$$

Now evaluate $g = \rho + \|\dot{\Psi}\|$ and finally (26). Matrix and transpose actions use the same reconstructed operator. One factors G and solves linear systems; an explicit inverse is unnecessary.

The response clock is more regular in approximation coordinates, but it is not automatically better at low order. At $P = 1$ the polynomial space is constant, so changing the coordinate alone leaves the physical constant-mode closure unchanged. Higher orders pay for Gram solves and derivative passes, and the theoretical constants can be large.

5 What is compressed?

Let H be hidden depth, n width, m training samples and P moment order. The comparison in Table 1 separates moving learned state from fixed source storage.

The learned action is matrix-free:

$$\widehat{W}^{(\ell)} v = W_0^{(\ell)} v - \frac{2}{nm} \sum_a U_{\ell a} G^{-1} H_{\ell a}^\top v \quad (\text{with the prefix correction in Variant B}),$$

and the transpose uses the same factors in reverse order. Storage is independent of elapsed training steps. The initialized operator remains uncompressed and still costs $O(n^2)$ per vector under a direct dense action. The current result is therefore compression of the *learned moving increment*, not total elimination of dense operators. Total stored memory can exceed a dense implementation if W_0 and all histories are materialized.

Table 1: Leading storage and direct-operation counts. A full batch applies each internal map to m vectors. Counts omit lower-order activations and first/readout layers.

	Dense gradient flow	Response-memory closure
Moving internal state	$(H - 1)n^2$	$2(H - 1)mnP$
Fixed internal source	included above	$(H - 1)n^2$ for all $W_0^{(\ell)}$
Learned rank/link	up to n	$\leq mP$ (A), $\leq m(P + 1)$ (B)
One learned action	$O(n^2)$	$O(nmP)$
One full-batch action	$O(mn^2)$	$O(nm^2P)$
Moment dilation	—	$O((H - 1)mnP)$
Variant-B overhead	—	$O(P^3 + (H - 1)mnP^2)$

Ignoring constants and the order threshold, Theorem 3.1 gives the sufficient scaling $P = O(\varepsilon^{-1})$ for trajectory error ε , while Theorem 4.1 gives $P = O(\varepsilon^{-1/2})$. Thus the proven moving-state scalings are respectively

$$O(Hmn\varepsilon^{-1}) \quad \text{and} \quad O(Hmn\varepsilon^{-1/2}).$$

These are approximation guarantees, not practical speed laws: C_T and $P_0(T)$ depend on the task, width, depth and horizon, and Variant B has larger per-step overhead.

5.1 Population interpretation and its precise limit

For fixed P , the neuron-indexed vectors can be viewed as particles sampling population fields indexed by the initialization source. This makes the closure a natural Monte Carlo implementation of a hybrid population/operator system: the response fields move, while the initialized random operator and its adjoint remain exact. The theorems above, however, keep n fixed and their constants may depend on n .

There is still a useful diagonal statement. Suppose an independent theorem establishes that dense width- n flows θ_n converge on $[0, T]$ to a population object Θ in a metric d , and suppose the observables entering d are controlled by Theorem 3.1 or Theorem 4.1. Choose any $n_j \rightarrow \infty$, then choose P_j large enough that the fixed-width closure error is at most $1/j$. The triangle inequality gives

$$d(\hat{\theta}_{n_j, P_j}, \Theta) \leq d(\hat{\theta}_{n_j, P_j}, \theta_{n_j}) + d(\theta_{n_j}, \Theta) \rightarrow 0.$$

This constructs a sequential approximation to the population flow. It does not prove that a fixed P works uniformly in width, that the W_0 action has a finite-population representation, or that the two limits commute. Rigorous multilayer mean-field and high-dimensional DMFT results exist in other regimes [Nguyen and Pham, 2023, Celentano et al., 2021]; matching their architectures, initializations and observables is a separate theorem.

6 Why this is not a frozen dictionary

A frozen dictionary chooses response directions at initialization and trains only their coefficients. It can be compact, but it cannot rotate its atoms to follow newly learned features. A trainable low-rank factorization allows more motion, yet its factor-gradient geometry is still different from the integrated rank-one updates of dense gradient flow.

Response memory keeps a low-rank learned correction at every instant, but its atoms are moments of the *current closure trajectory*. The loop is

$$\text{moments} \longrightarrow \widehat{W} \longrightarrow \text{current forward/backward responses} \longrightarrow \text{moments}.$$

The basis in history coordinate is fixed; the neuron vectors multiplying that basis coevolve nonlinearly. This is why the method can represent substantial feature learning without being a frozen tangent model.

On the hard two-outlier circle task at width 2048, the current study reports RMS discrepancies 1.100 and 0.756 for frozen derivative dictionaries with 18 and 72 stored vectors, and 0.896 for a trainable 18-vector derivative dictionary. Response memory gives 0.07295 at $P = 3$ (48 vectors) and 0.00466 at $P = 7$ (112 vectors). In a more controlled comparison retaining the same W_0 , response moments beat a trained factorization $W_0 + AB$ in all 29 valid task/order/seed cells; on the hard task at rank 56 the errors were 0.004663 versus 0.516210 and 0.251026. These observations isolate a mechanistic difference from those baselines. They do not prove superiority over every adaptive dictionary or at matched total memory.

7 Empirical evidence

The experiments use finite networks and numerical time integration. Circle scores compare predictors on 8192 passive angles; MNIST scores use held-out images. Most comparisons match each model at its own first training-MSE 0.001 crossing. They therefore test agreement of fitted functions, while the theorems control same-time trajectories. The panels are selected finite-instance studies, generally with one initialization, and do not estimate the asymptotic theorem constants. The three figures below use the ordinary activity clock.

Variant B has also been implemented with batched Gram solves and directional response derivatives. In a matched width-2048 comparison on three circle tasks with tanh depth two and GELU depth three, all 49 dense/closure fits reached training RMS at most 0.05; its $P = 3$ predictor discrepancy was below 0.05 in five of six principal groups. The GELU alternating-quadrant exception improved from 1.245 to 0.112 after halving the time step. In a broader 82-case sweep, the fitted-pair median whole-query discrepancies for $P = 1, 2, 3$ were 0.0427, 0.0189, 0.00680, but several fitted alternating-label cases remained poor. These runs establish implementability and useful low-order cases. They do not empirically certify the $O(P^{-2})$ rate or show that Variant B uniformly beats Variant A.

Own-endpoint functions on the circle
Width 2048; each model at its own training-MSE 0.001 endpoint; closure retains dense W0

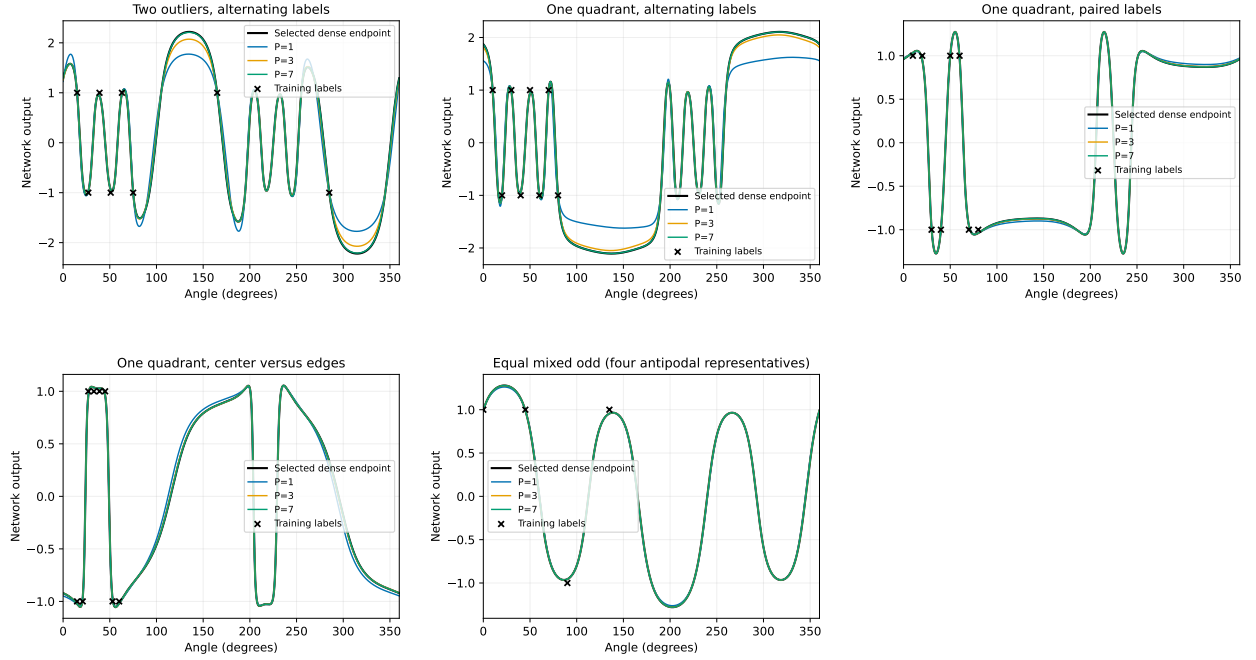


Figure 1: Two-hidden-layer tanh networks of width 2048. The black curve is the dense fitted function on the whole circle; colored curves are ordinary-clock closures. Training labels are marked by crosses. Increasing order reproduces the dense predictor between and far away from training inputs.

Across the five tasks in Figure 1, the measured RMS errors are:

Task	$P = 1$	$P = 3$	$P = 7$
Two outliers, alternating	0.2363	0.0730	0.00466
Quadrant, alternating	0.3414	0.0457	0.00165
Quadrant, paired	0.0193	0.00282	≈ 0.00012
Quadrant, center/edges	0.0456	0.00487	0.00130
Equally spaced, mixed odd	0.0124	0.000520	≈ 0.00001

These tasks include rapid alternation and large unlabeled arcs, so training-point agreement alone would not force the observed whole-circle match.

Functions on the circle at matched training loss

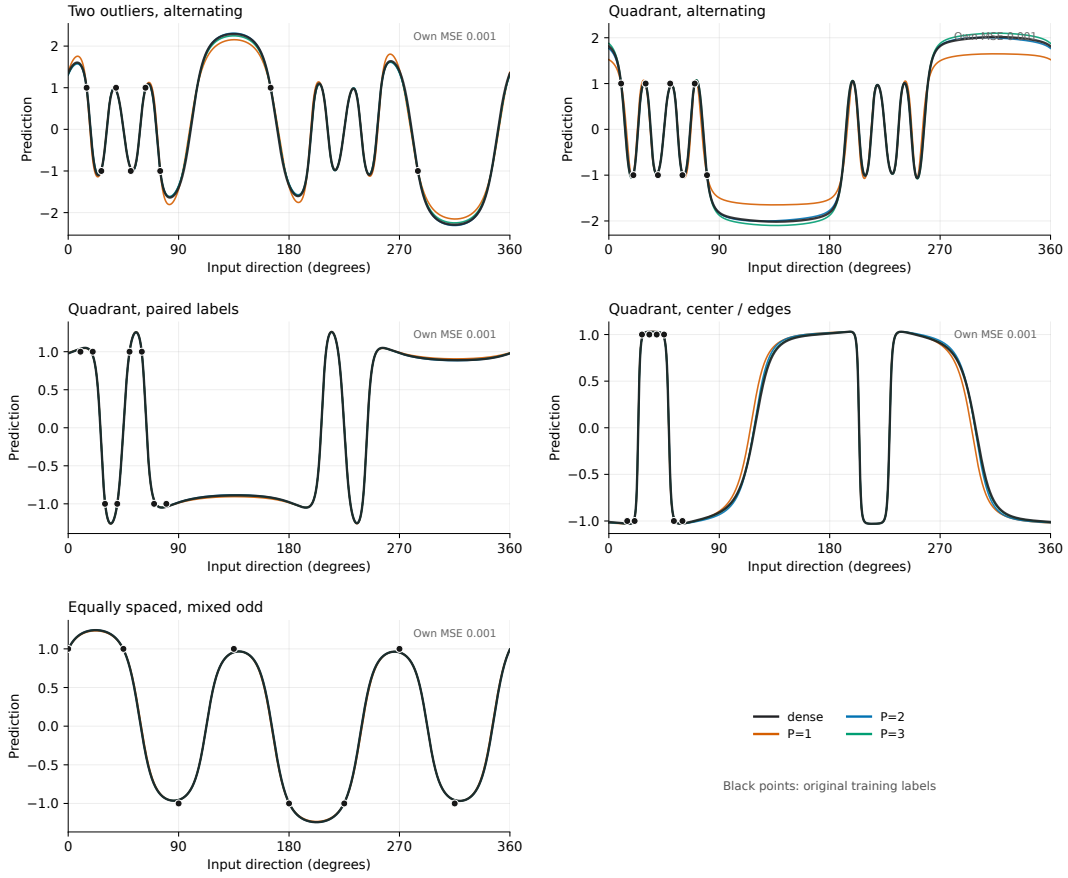
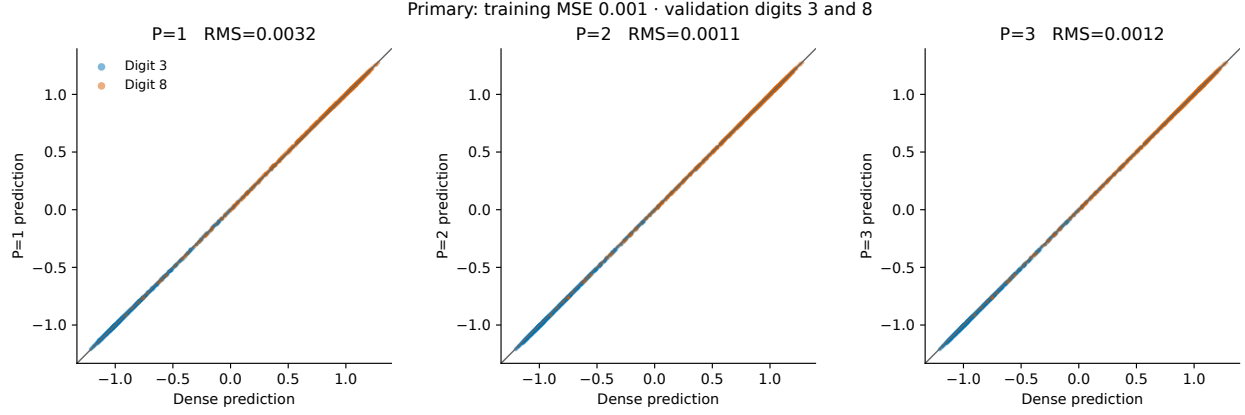
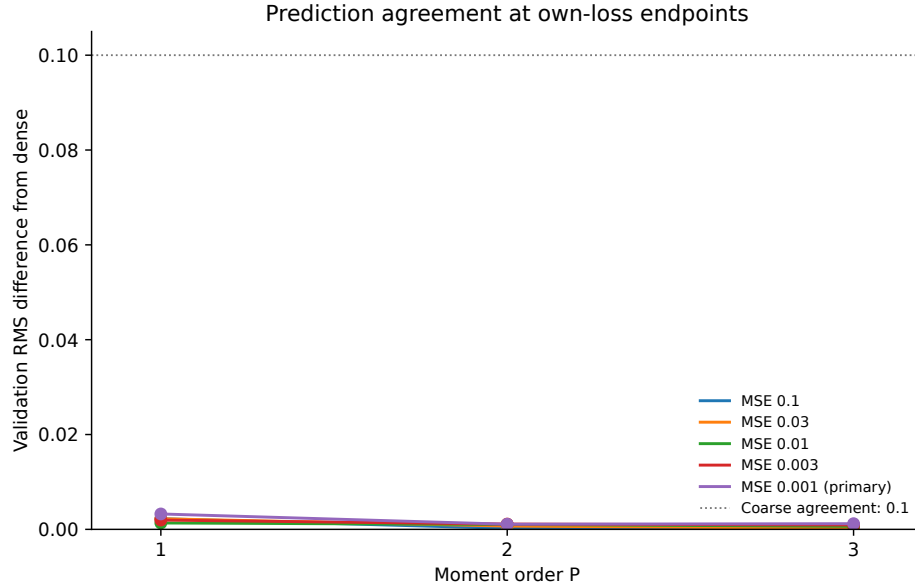


Figure 2: Three-hidden-layer tanh networks of width 4096, compared at matched training loss. Across the five tasks, $P = 3$ whole-circle RMS discrepancy ranges from 2.82×10^{-4} to 5.85×10^{-2} . The order trend is not pointwise monotone: $P = 2$ beats $P = 3$ on the two alternating tasks.

The depth-three circle campaign also measures nontrivial hidden-feature motion, so the agreement is not explained by a frozen-feature limit. Experiments with GELU and sigmoid show that the construction is not tanh-specific, but accuracy is task dependent: the reported GELU hard-outlier $P = 3$ error is 0.02485, whereas harder GELU/sigmoid cases are near 0.26 and 0.24. ReLU and SELU runs are useful empirical probes but fall outside the smooth ODE theorem at their kinks; selected ReLU flows can even be nonunique at exceptional initial states.



(a) Closure versus dense predictions on 1984 held-out images.



Bars: observed refinement margins, not confidence intervals. Open circles: numerical gate failed.

(b) RMS discrepancy across several matched-loss endpoints.

Figure 3: MNIST 3-versus-8, 100 training images, width 4096. At training MSE 0.001, $P = 1, 2, 3$ have held-out RMS discrepancies 0.00324756, 0.00108443, 0.00119922. All declared numerical-refinement gates passed; the slight $P = 3$ worsening shows that autonomous feedback need not make realized error monotone in P .

For this MNIST panel the learned rank bounds are 100, 200, 300. The reported learned-increment storage is 6.25, 12.5 and 18.75 MiB, versus 128 MiB for the unrestricted learned internal correction. The closure also retains the 128 MiB initialized matrix, so its minimum total state is not smaller than a dense implementation storing only current weights. The measured adaptive closure integration was also slower in this panel. The empirical achievement is fidelity with a compressed moving update, not an end-to-end speedup claim.

8 Limitations and next questions

The main positive result is a fixed-order family of autonomous systems whose state does not grow with elapsed training time and whose physical trajectories converge to dense gradient flow as $P \rightarrow \infty$ on every prescribed finite horizon. Several boundaries matter.

- The initialized random matrices and their transpose actions are retained. Compressing that source while preserving matrix reuse is still open.
- Constants are not uniform in width, depth, sample count or horizon. A direct fixed- P population theorem and a CLT-style finite-width error theory remain future work.
- Neither theorem gives one order valid for all $t \geq 0$. Uniform all-time tracking would require bounded accumulated activity and nonlinear stability with finite total amplification.
- Smooth activations such as tanh, sigmoid, arctangent, GELU, softplus, SiLU, sine and polynomials fall within the stated local smoothness classes. Literal ReLU and standard SELU do not. Their discontinuous selected backward fields require nonsmooth dynamics and a clock that charges jump variation.
- A finite scalar aggregate truncation is a second approximation, beyond history truncation. Differentiating aggregates gives a principled hierarchy, but an economical width-uniform scalar closure is not established here.
- Matching the dense predictor is different from predicting the target well. Generalization requires an additional argument about the function selected by training.

The response-memory construction nevertheless gives a concrete answer to the paper’s question. Exact dynamics can hide a growing history inside a formal population or operator description. Approximate dynamics can replace that history by finitely many evolving coordinates per neuron, retain nonlinear feature feedback, and admit a genuine order-versus-trajectory-error theorem. That smaller state is a new lens on the learned part of deep training, even before the remaining random operator is compressed.

References

- Michael Celentano, Chen Cheng, and Andrea Montanari. The high-dimensional asymptotics of first order methods with random data, 2021. URL <https://arxiv.org/abs/2112.07572>.
- Lénaïc Chizat and Francis Bach. On the global convergence of gradient descent for over-parameterized models using optimal transport. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Phan-Minh Nguyen and Huy Tuan Pham. A rigorous framework for the mean field limit of multilayer neural networks. *Mathematical Statistics and Learning*, 6(3/4):201–357, 2023. doi: 10.4171/MSL/42.

Greg Yang and Edward J. Hu. Tensor programs IV: Feature learning in infinite-width neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 11727–11737. PMLR, 2021. URL <https://proceedings.mlr.press/v139/yang21c.html>.